Faculty of Engineering and Computer Science
Expectations of Originality

This form sets out the requirements for originality for work submitted by students in the Faculty of Engineering and Computer Science. Submissions such as assignments, lab reports, project reports, computer programs and take-home exams must conform to the requirements stated on this form and to the Academic Code of Conduct. The course outline may stipulate additional requirements for the course.

1. Your submissions must be your own original work. Group submissions must be the original work of the students in the group.
2. Direct quotations must not exceed 5% of the content of a report, must be enclosed in quotation marks, and must be attributed to the source by a numerical reference citation[1]. Note that engineering reports rarely contain direct quotations.
3. Material paraphrased or taken from a source must be attributed to the source by a numerical reference citation.
4. Text that is inserted from a web site must be enclosed in quotation marks and attributed to the web site by numerical reference citation.
5. Drawings, diagrams, photos, maps or other visual material taken from a source must be attributed to that source by a numerical reference citation.
6. No part of any assignment, lab report or project report submitted for this course can be submitted for any other course.
7. In preparing your submissions, the work of other past or present students cannot be consulted, used, copied, paraphrased or relied upon in any manner whatsoever.
8. Your submissions must consist entirely of your own or your group's ideas, observations, calculations, information and conclusions, except for statements attributed to sources by numerical citation.
9. Your submissions cannot be edited or revised by any other student.
10. For lab reports, the data must be obtained from your own or your lab group's experimental work.
11. For software, the code must be composed by you or by the group submitting the work, except for code that is attributed to its sources by numerical reference.

You must write one of the following statements on each piece of work that you submit:
For individual work: **"I certify that this submission is my original work and meets the Faculty's Expectations of Originality",** with your signature, I.D. #, and the date.
For group work: **"We certify that this submission is the original work of members of the group and meets the Faculty's Expectations of Originality",** with the signatures and I.D. #s of all the team members and the date.

A signed copy of this form must be submitted to the instructor at the beginning of the semester in each course.

I certify that I have read the requirements set out on this form, and that I am aware of these requirements. I certify that all the work I will submit for this course will comply with these requirements and with additional requirements stated in the course outline.

Course Number:	COEN 317	Instructor:	Dr. Fadi El-Hassan
Name:	Simon Guindon	I.D. #	40058301
Signature:	Simon Guindon	Date:	23-09-2025

[1] Rules for reference citation can be found in "Form and Style" by Patrich MacDonagh and Jack Bordan, fourth edition, May, 2000, available at http://www.encs.concordia.ca/scs/Forms/Form&Style.pdf.
Approved by the ENCS Faculty Council February 10, 2012

assignment 1 COEN 317 Simon Guindon 40058301

Q1 Address bus 22 bits → 2^{22} addresses (bit addressable)

For kilo bytes: $\frac{2^{22}}{2^3} \cdot \frac{1}{2^{10}} = 2^9$ KB = 512 KB

↑ bytes ↑ kilo

Q2 Address bus 22 bits → byte addressable $\frac{2^{22}}{2^{10}}$ bytes $= 2^{12}$ KB

↑ kilo 4096 KB

Q3 Address bus 27 bits → byte addressable $\frac{2^{27} \text{ bytes}}{2^{10}} \cdot \frac{1}{2^{10}} = 2^7$ MB

↑ kilo ↑ mega = 128 MB

Q4 28 bits (half word addressable)

$2^{28} \cdot 2$ (since 2 bytes in half word) $= 2^{29}$ bytes

$\frac{2^{29}}{2^{10}} \cdot \frac{1}{2^{10}} = 2^9$ MB = 512 MB

Q5 25 bit address bus → $2^{25} \times 4$ bytes $= 2^{27}$ bytes

$\frac{2^{27}}{2^{10}} \frac{1}{2^{10}} = 2^7 = 128$ MB

Q6 a) the cell of the instruction memory should be 16 bits to fetch one instruction per cycle

the cell of the data memory should be 8 bits to fetch one register per instruction

b) 1 MB = 2^{20} bytes and each register is one byte → 2^{20} locations

so we need 20-bits address bus from 0x00000 to 0xFFFFF

c) 1 kB = 2^{10} bytes but each instruction is 16 bits (2 bytes) so $\frac{2^{10}}{2} = 2^9$ ⇒ we need 9 bits address bus

0x000 to 0x1FF

7) $\frac{7\times10^6 \text{ CC}}{2.4\times10^{-3} \text{ s}} = 2.92\times10^8 \text{ Hz}$ for Comp A

Comp B $= 3.5\times 2.92\times10^8$ Hz

Comp B $= 1.022\times10^9$ Hz

$1.022\times10^9 \text{ Hz} \times 8\times10^{-3} = \boxed{8.176\times10^6 \text{ CC}}$

8) a) Since the instructions are 2 bytes long and the memory is byte addressable, we need to increment PC by two to get the next instruction

b) Four instructions would be adding 8 to the PC

$$\begin{array}{r} 0000\,CEFF \\ +\quad 8 \\ \hline 0000\,CF07 \end{array}$$

note i dont know why there is a H in the question (Hexadecimal?)

c) 0x190D

0001 1001 0001 0000 1111 01

can only be the third instruction

ADDS R6, R5, R3

adds register value instead of immediate value

0x1D0D

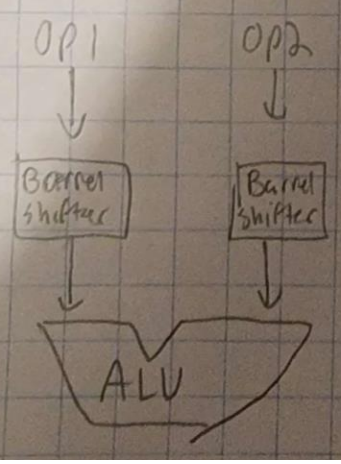
0001 1101 0001 0000 1111 01

would have been 2nd instruction

ADDS R4, R2, #6

adds immediate value instead of register value

9) Usually we would only need a barrel shifter for the 2nd operand, but now we need to support barrel shift for the first operand



now upon the new instruction being used the shift will occur in the first operand

10a) 32 bytes register instruction 16 bits

16 bits instruction

15:10	9:7	6:4	3:0
6 bits	3 bits	3 bits	4 bits
15 – 11	7	3	0
Opcode for 64 instruction	Rds (6 register)	Rs (6 register)	Immediate value

b)

31:26	25:21	20:16	15:11	10:0
31 30 29 28 27 26	25 24 23 22	21 20 19 18 17 16	15 14 13 12 11 10	9 8 7 6 5 4 3 2 1 0
6 bits for 64 instructions	5 bits for 32 register	5 bits for 32 register	5 bits for 32 register	2048 posible (11 bits) values for immediate

c)

31:26	25:21	20:16	
6 bits	5 bits	5 bits	
31 30 29 28 27 26	25 24 23 22	21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
6 bits for 64 instruction	5 bits for 32 registers RDS	5 bits for source register	2^{16} for immediate value

11) Adds R1, R2, R3 where 8 bits registers R2 = 255
R3 = 2

Since 255 + 2 = 257, we get C flag from the addition

ADDS R1, R2, R3 where 8 bits using signed (most significant bits is +/−)

R2 = 0x7F (+127) R3 = 0x01 gives 0x80 → −1 instead of +128

so overflow occured, set V flag